

M41 — Printable QR sheet for the instructor

Phase: Phase 7 follow-on (post-M40, operator-feature request) **Status:** **DONE** **Blocked by:** M2 (Encounter dataclass + room_code), M22 (Per-Patient Console), M31 (per-encounter QR card with all 3 station URLs), M36 (Nursing Station QR pattern) **Blocks:** none **Estimated effort:** 0.5 day

1. Purpose

Operator feature request:

"The instructor should have the option of printing out a page of QR codes for related to each encounter, for any devices, characters, records management. The instructor should be able to select printing all the QR codes for each encounter or select a single encounter. Each QR code should be clearly labeled and if any codes are needed to sign in and join that should also be printed at the top. The Patient Character should also be on the header and the header should have a title 'Training Bridge MedSim-VRAI'."

Implementation: new `/portal/control/qr_print` route that renders a print-friendly sheet. One page per encounter; optional `?encounter_id=...` query param scopes to a single bed.

Each page carries:

1. **Title bar:** `"Training Bridge MedSim-VRAI"` in large brand color, with a `"Multi-patient simulation — station join sheet"` subtitle.
2. **Patient character banner:** the persona's display name + id + role in a warm-yellow banner so it's the first thing anyone reading the sheet sees.
3. **Sign-in codes block:** the room code + the bed's join code in a typed-friendly monospace, plus the EHR system id when configured. These are what someone manually types when the QR can't be scanned.
4. **QR grid** — four blocks (2×2): Chat station, EHR station, Device station, Nursing Station. Each block has a label, a one-line description, the QR image (`/api/qr.svg?data=...`), and the URL printed underneath in monospace.
5. **Page-break-after** between encounters so each bed gets its own sheet of paper.

Two launch buttons:

- **Multi-Patient Control header** → **Print QR codes** →

`/portal/control/qr_print` (all encounters, new tab).

- **Per-Patient Console QR card footer** → **Print QR codes for**

this encounter → `/portal/control/qr_print?encounter_id={id}` (just this bed, new tab).

The new tab carries a **Print** action button at the top. The operator clicks Print → browser print dialog → paper or save-as- PDF. The action bar is hidden via `@media print` so it doesn't land on the printed page.

2. Structure

Files touched:

- `portal/templates/qr_print.html` (new) — full standalone HTML

page (does NOT extend `base.html` — we don't want the portal nav on the printed sheet). Inline CSS sets `@page` size + margins, a screen-only off-page background, `page-break-after per .qr-page`, and a `@media print` block that strips the off-page chrome.

- `portal/server.py` — new route

GET /portal/control/q_r_print (instructor-only via Depends(auth.require_vault)). Optional encounter_id query param. Resolves the active room, filters out private-clone per-student clones (template encounters are what the operator cares about for QR distribution), and hydrates each encounter with its patient persona's display name via library.get_persona. Renders the template with room_code, encounters, base_url, scope_label. Handles "no active room" gracefully (empty notice).

- portal/templates/control_room.html — new `<a id="btn-gr-print">`

in the room top bar, between the scene-inject button and the cohort-debrief button.

- [portal/templates/encounter_console.html](#) — new

 in the QR-codes card's footer (scoped to this encounter via ?encounter_id={{ encounter.id }}).

- portal/static/control_room.css — .room-topbar a#btn-gr-print

styled to match the other `.secondary` buttons.

- portal/static/encounter_console.css — .qr-print-link styled

as a primary blue pill button.

No backend schema change. No new dataclass field.

3. Uses

3.1 Operator print-all flow

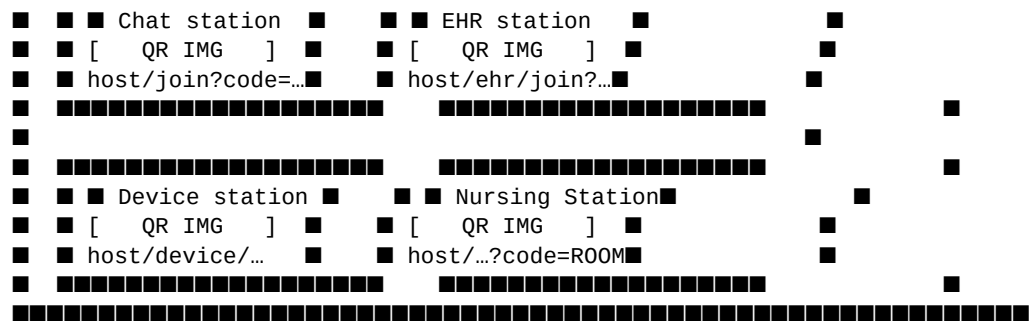
1. Operator opens /portal/room (Multi-Patient Control). 2. Clicks **Print QR codes** in the header. 3. New tab opens at /portal/control/qrcode_print. 4. Each encounter renders as one printable page with: - "Training Bridge MedSim-VRAI" title bar. - Patient character banner (e.g. "Patient character: Mr. Hayes (P-014 - Hyperactive Delirium)"). - Sign-in codes: room code 2632B8, bed join code 7J7AVM, EHR system helix. - Four QR blocks: Chat / EHR / Device / Nursing Station. 5. Operator hits **Print** (top of page) or Cmd/Ctrl+P. 6. Browser opens its print dialog. Operator chooses printer or saves as PDF. Each encounter is one page of the output.

3.2 Per-encounter print flow

1. Operator opens /portal/room/encounter/{id} (Per-Patient Console). 2. Scrolls to the QR-codes card. 3. Clicks **Print QR codes for this encounter** in the footer. 4. New tab opens at /portal/control/qr_print?encounter_id={id}. 5. One page renders — just this bed's encounter. 6. Same Print flow.

3.3 What's on each printed page

[illegible]



The Nursing Station block is tinted green and its sublabel says **"Supervisor view — uses the room code, not the bed join code"** so the operator handing the sheet to a nursing-station tablet doesn't type the wrong code.

4. Functions (exported API surface)

Symbol	Where	Purpose
GET /portal/control/qr_print	portal/server.py	Renders the print sheet. Optional ?encounter_id=<id> scopes to one bed. 404 on unknown encounter; renders an empty notice when there's no active room.
portal/templates/qr_print.html	(new)	Standalone HTML — doesn't extend base.html (no portal nav on printed paper). Inline CSS with @page + @media print + page-break-after.

5. Limitations

- **Private-clone clones are skipped.** When ``chart_mode == "private_clone" and cloned_from_id`` is set, the encounter is a per-student clone of a template. The print sheet shows only the template encounter (the one the operator authored). Clones inherit the template's QR codes anyway. Acceptable; matches the M13 model.
- **Persona name comes from `library.get_persona()`.** If the persona ID is not in the canonical 24-persona library (defensive edge), the name is empty and the banner shows just the id. Same pattern as the M33 voices endpoint.
- **Page size hard-coded to US Letter portrait.** International operators wanting A4 can change the print dialog's "Paper size" in the browser; `@page size: letter portrait` is the *suggested* default, not enforced. A future M42 could read the user-agent's locale.
- **Each page is one encounter.** With a 10-encounter room, that's 10 sheets. Could fit two encounters per page on a denser layout, but the user explicitly asked for clear labels + per- encounter QR sets, so one-encounter-per-page is the cleaner default.
- **QR codes are SVG via `/api/qr.svg?data=...`** Same endpoint as M31's on-screen QRs. SVG prints crisply at any resolution. If the printer rasterizes oddly, a future M42 could switch to PNG.
- **No CSV export of the codes themselves.** Operators wanting a

digital ledger of room/join codes can save the print page as PDF — the codes are in plain text in the headers. CSV is out of scope for v7.0.

6. Test status

Test file	Asserts	Status	Last run
tests/v7/test_qr_print_sheet.py::test_qr_print_route_returns_branded_print_page	Page contains brand title, print action, back link, page-break-after, @media print	PASS	2026-05-27
...::test_qr_print_route_renders_all_encounters_by_default	All encounters' join codes present + scope label says "all N encounters"	PASS	2026-05-27
...::test_qr_print_route_scoped_to_single_encounter	With ?encounter_id, only that bed's join code is in the output	PASS	2026-05-27
...::test_qr_print_route_404_on_unknown_encounter	Unknown encounter id → 404	PASS	2026-05-27
...::test_qr_print_route_handles_no_active_room	No active room → 200 with empty notice (no 500)	PASS	2026-05-27
...::test_qr_print_page_carries_patient_character_in_header	"Mr. Hayes" + persona ids appear; "Patient character" label appears	PASS	2026-05-27
...::test_qr_print_page_carries_room_and_join_codes	Both room code + bed join code rendered with "Room code" + "Bed join code" labels	PASS	2026-05-27
...::test_qr_print_page_renders_all_four_station_qrs_per_encounter	Chat / EHR / Device / Nursing labels + URLs all present; Nursing uses ROOM code	PASS	2026-05-27
...::test_qr_print_page_qr_imgs_use_api_qr_svg_endpoint	QR tags target /api/qr.svg?data=	PASS	2026-05-27
...::test_multi_patient_control_header_has_print_qr_button	/portal/room has #btn-qr-print anchor → /portal/control/qr_print with target=_blank	PASS	2026-05-27
...::test_encounter_console_has_per_encounter_print_link	Per-Patient Console QR card has .qr-print-link → /portal/control/qr_print?encounter_id=<id>	PASS	2026-05-27
Full v7 suite	274 passed, 1 skipped (M20 Playwright skip — unchanged)	PASS	2026-05-27

7. Change list

Date	Author	Change	Files
2026-05-27	claude-code	Initial M41 implementation: print template + /portal/control/qr_print route + two launch buttons; 11 new tests	portal/templates/qr_print.html (new), portal/server.py, portal/templates/control_room.html, portal/templates/encounter_console.html, portal/static/control_room.css, portal/static/encounter_console.css, tests/v7/test_qr_print_sheet.py (new)

8. Open questions / known issues

- **Print as PDF and attach to course materials?** The browser's

"Save as PDF" already does this. If operators want a server- rendered PDF download (so they can email or attach without going through the browser), a future M42 could add a `?format= pdf` variant using `weasyprint` (already a dependency for the module-guide PDF renderer).

- **Future-device QR codes** (call bell, bed alarm, code blue, fire

alarm)?** M29 future-device stubs press via the encounter console's `/api/encounter/{id}/future_device/{kind}/press` — there's no separate join URL for them. If a future module adds hardware bindings (real call bells on real devices), the print sheet should add a 5th block.

- **Operator-customized header.** Some classrooms may want the

school logo + course code in the header. Hard-coded today; a future M42 could read a customization JSON from `~/.medsim/v7/ brand.json`.

- **Localization.** Title + labels are English only. Acceptable

for v7.0; a future M42 could externalize.

Render this guide to PDF with `python docs/module_guides/render_pdfs.py`.